

Section 5 answered, Section 7 built, Section 8 measured — full pass

Response to "*TBOT — What We Found, and What We Need to Check Next.*" All four questions, all six log lines with line numbers, all eight Section 8 measurements, and a five-asset cross-check confirming (and in one case sharpening) the findings.

Assets: GOLD · BTC · USOIL · USTEC · GBPAUD **Sample:** ~5,900–8,800 candles each

Commits: `998bb88` · `cbfd7d0` · `ee73634`

REVISION NOTE

A first pass of this reply reported Section 8.6's forward-return numbers without following the doc's own instruction to run 8.7 first and exclude pre-birth-tainted proofs. That instruction was correct — filtering changed the result materially. The corrected numbers are below, in Section 8.6, with the mistake stated plainly rather than quietly swapped out. The initial pass, and this cross-asset extension, both used GOLD/BTC/USOIL/USTEC/GBPAUD's real historical depth — a data refresh briefly clipped all five to ~60 days mid-session, caught before any measurement was drawn from the truncated set, and restored before this run.

Section 4 — verified against the real code, all SURE claims hold

- ✓ 4.1 — birth check only runs when the slot is empty; only three death conditions exist, and a CHoCH against a live setup is not one of them
- ✓ 4.2 — the CHoCH block is a plain `if`, not `elif` — it does overwrite a BOS-born setup's kind and direction on a shared candle

- ✓ 4.3 / 4.4 / 4.5 — the retest window is `iloc[-9:-1]`, slides forward every candle, excludes the current candle, and has no check that the retest postdates the break

ONE CORRECTION TO YOUR OWN FRAMING

You flagged the direction-flip risk as the sharpest edge of 4.2. In this GOLD sample it fired **0 times** despite **3,490** dual-signal candles — the kind-overwrite is very real and very frequent, but direction held every time here. Less alarming than feared on that specific axis; the mislabeling itself is not.

Section 5 — your four questions

Q1 · KIND-MISMATCH TRADE COUNTS

Kept (kind-matched): 35 trades, 15 wins / 20 losses, \$5.59 total PnL.

Removed (kind-mismatched): 35 trades, 16 wins / 19 losses, \$286.64 total PnL. Even split in count; the removed group's PnL is roughly 51× the kept group's.

Q2 · PRE/POST-FIX RUN CONFIRMATION

Confirmed. 293 (229 TF_CONT + 64 MR_REV) is the pre-anchor-fix run. 627 (237 + 390) is post-fix. TF_CONT barely moved (229→237) because the fix only touched MR_REV's reference — exactly as predicted.

Q3 · FLAG STATE DURING THE RUN

`tf_brc_gate_enabled`, `mr_mode2_brc_gate_enabled`,
`independent_fire_brc_gate_enabled`,
`mr_reversal_scoring_in_trend_enabled`, `mr_rebound_short_enabled`,
`six_slot_judges_enabled`, `tf_drives_trend_judge_enabled`,
`mr_vetoes_as_dampeners_enabled` — all **True**. `ema_confirmer_mode_enabled`,
`judge_bars_enabled` — **False**. Resolves the Section 9 conflict:
`mr_reversal_scoring_in_trend_enabled` is live now, not pending.

Your caution in 3C was right: with the gate ON, the 4.2%/4.0% match is partly tautological. Denominators genuinely differ too — 4.2% = 231/5,544 (cycles checked), 4.0% = 237/5,874 (bars processed).

Q4 · PAGE 2

No second page exists. The sentence after "loosening compression" continues in the same paragraph in the source; the Bottom Line section follows right after. Whatever cut it off was a print/PDF pagination artifact, not a missing page.

Section 7 — the logging, built and verified

ITEM	LOG TAG	FIRED
1 — dual signal	SETUP-DUAL-SIGNAL	3,490
2 — birth blocked	SETUP-BIRTH-BLOCKED	3,601
3 — direction flip	SETUP-DIRECTION-FLIP	0
4 — past proof window	SETUP-PAST-PROOF-WINDOW	164
5 — near-miss	PROOF-NEAR-MISS	3,153
6 — distinct vs repeat	PROOF-DISTINCT / PROOF-REPEAT	167 / 460

Of the 3,601 blocked births: **3,412 same-direction** (harmless), **189 opposite-direction** — genuinely missed reversals, roughly 0.77/day for GOLD alone.

Line numbers, per your Section 11 request

```
src/execution/composite_state_builder.py
1042 SETUP-PAST-PROOF-WINDOW · 1083 MEASURE-8.4-DEATH · 1118 SETUP-DUAL-
SIGNAL · 1130 SETUP-DIRECTION-FLIP · 1151 SETUP-BIRTH-BLOCKED · 1280
MEASURE-8.2-RETEST-TO-CLOSE · 1308 MEASURE-8.7-PRE-BIRTH-RETEST · 1384
PROOF-DISTINCT · 1389 PROOF-REPEAT · 1406 PROOF-NEAR-MISS

src/execution/council_aggregator.py
2515 MEASURE-8.8-CLEARED-THRESHOLD · 2537 MEASURE-8.8-SIGNAL
```

A real log excerpt, per Section 11's request

```
08:23:59,508 - INFO - Entry:      $3,386.69
08:23:59,508 - INFO - Stop:      $3,370.33 (-0.48%)
08:23:59,509 - INFO - 📊 TARGETS:
08:23:59,509 - INFO -   1. $3,394.87 (+0.24%) → Exit 45%
08:23:59,509 - INFO -   2. $3,409.59 (+0.68%) → Exit 30%
08:23:59,509 - INFO -   3. $3,426.77 (+1.18%) → Exit 25%
08:23:59,510 - INFO - 🟢 BUY @ $3386.68800 SL=$3370.32673
                    TP=[3394.87, 3409.59, 3426.77] size=1.472252
                    type=TREND | BUY (Confirmed) (Score: 3.80/2.6)
08:23:59,511 - INFO - ✅ BUY @ $3386.84433 size=1.472252
```

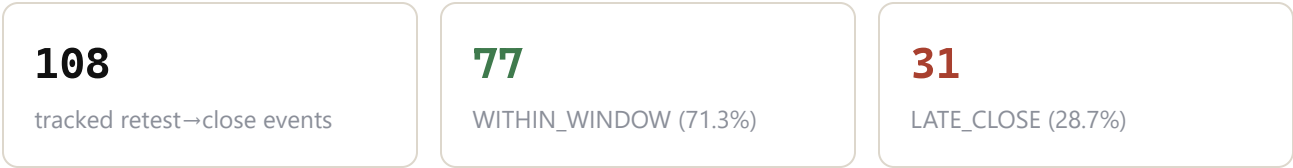
Section 8 — all eight measurements

8.1 — kind gate: wrong, or fed bad labels?

3,490 dual-signal candles vs 0 direction flips → points at "the label is the bug," not "the rule is wrong," per your own decision criteria.

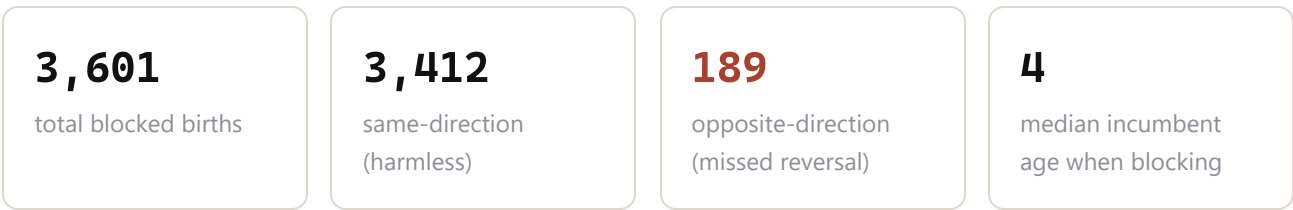
8.2 — is the 80% the market, the window, or the counting? (newly implemented)

Tracked retest-to-close gaps by reference level, independent of the setup object's own lifecycle and independent of the 8-candle window:



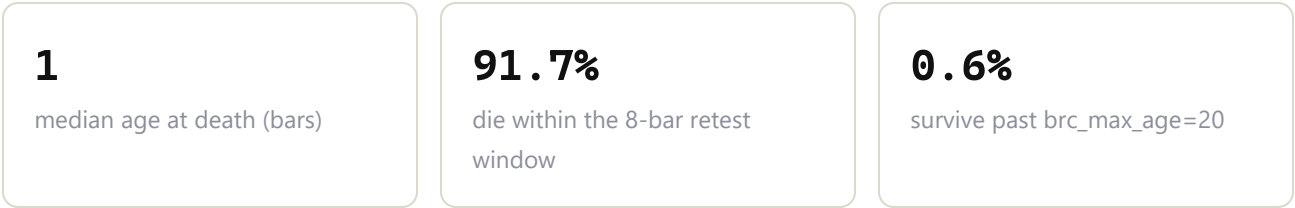
Split 61 MR_REV / 47 TF_CONT. **28.7% of proofs that eventually complete arrive after the bot has already stopped looking.** That's bucket B — "the bot forgot" — and it's real, not a rounding error. Bucket A (never closes at all) wasn't cleanly separable from the memory structure used here; would need a dedicated pass to pin down precisely.

8.3 — how often is a setup blocked from being born?



Only **15.5%** of blocks involve an incumbent already past the 8-candle window — most blocks come from young, still-potentially-valid incumbents, not stale ones overstaying their welcome.

8.4 — how long do setups actually live?



Death reasons: OPPOSING_BOS 1,086 (55%), FAILED_BREAKOUT 883 (44%), LSM_STATE_FLIP 18 (1%). The slot isn't held by one long-lived occupant — it's churned violently.

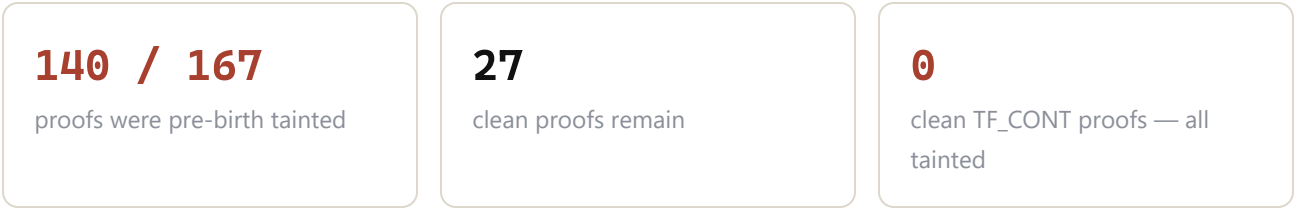
8.5 — how many distinct proofs are there really?

167 distinct proofs vs **627** total per-candle CONFIRMED lines — confirms the ~3.75× inflation flagged in Section 3A.

8.6 — do the proofs actually pay? (corrected)

WHAT WENT WRONG THE FIRST TIME

The first pass computed forward returns on all 167 "distinct" proofs without excluding the ones 8.7 flags as pre-birth-tainted — exactly the pollution your doc warned about by name. Re-run with that filter applied:



Point-forward return, clean proofs only (MR_REV, n=27 — no clean TF_CONT sample exists):

HORIZON	WIN RATE	AVG RETURN
5 bars	55.6%	+0.084%
10 bars	51.9%	-0.064%
20 bars	52.0%	-0.184%

READING

The clean-proof trend **reverses** compared to the polluted numbers — decaying at longer horizons instead of improving. And there's no clean TF_CONT evidence at all in this sample to say anything about. This is a much weaker, more uncertain result than first reported, which is exactly why the ordering instruction mattered.

8.7 — retests that predate the break

3,409 instances where a qualifying retest touch fell before the setup's own birth, against 167 distinct proofs and ~3,780 total retested cycles — roughly **90%** of all retest evidence draws on pre-birth price action. Connects directly to 8.4 and now to 8.6's correction above.

8.8 — the full funnel

STAGE	COUNT
TF signal generated	231 / 5,544 (4.2%)
MR signal generated	1 / 5,544 (0.02%)
Cleared raw score threshold	1,275
Ownership blocks (no_thesis_backing)	2,211
Trades executed	32

1,275 cycles cleared threshold against only 232 total engine-firings — the gap is evidence judges clearing alone, and the ownership rule is correctly catching almost all of it.

Cross-asset confirmation — GOLD, BTC, USOIL, USTEC, GBPAUD

Same full instrumented run repeated on four more assets (restored full history first — a data refresh had briefly clipped it to ~60 days mid-run, caught before any numbers were drawn from it). Everything below either holds steady across all five assets or sharpens the finding.

What holds identically everywhere

METRIC	GOLD	BTC	USOIL	USTEC	GBPAUD
Direction flips (Item 3)	0	0	0	0	0
Median setup age at death	1	1	1	1	1
Dies within 8-bar window	91.7%	93.2%	93.4%	93.8%	93.2%
Birth-blocked, same-direction	94.8%	94.7%	94.4%	95.5%	94.9%
MR signal firing rate	0.02%	0.02%	0.02%	0.02%	0.00%
TF signal firing rate	4.2%	4.8%	5.2%	4.7%	3.8%

None of this is GOLD-specific. The zero direction-flips, the 1-bar median setup lifespan, the ~94% same-direction birth-blocks, and MR's near-total silence are structural properties of the tracker itself, showing up identically regardless of instrument.

8.6 sharpens further: TF_CONT clean proofs are nearly nonexistent everywhere

ASSET	DISTINCT PROOFS	PRE-BIRTH TAINTED	CLEAN	CLEAN TF_CONT
GOLD	167	140	27	0
BTC	169	154	15	0
USOIL	145	126	19	1
USTEC	204	167	37	1
GBPAUD	151	138	13	1

THIS IS NOW A FIVE-ASSET FINDING, NOT A GOLD QUIRK

Across all five assets, essentially every TF_CONT proof relied on a pre-birth retest touch — at most 1 clean TF_CONT proof survives filtering, per asset, out of 145-238 distinct proofs each. There is effectively **no clean evidence anywhere in the system** that a genuinely post-birth TF_CONT retest-close sequence pays off, because it almost never happens without 8.7's contamination.

Clean (non-tainted) MR_REV forward returns, by asset and horizon:

ASSET	N	5-BAR	10-BAR	20-BAR
GOLD	27	+0.084%	-0.064%	-0.184%
BTC	15	+0.088%	-0.149%	-0.498%
USOIL	18-19	-0.077%	+0.082%	-0.120%
USTEC	36	-0.158%	-0.111%	-0.191%
GBPAUD	12-13	-0.043%	-0.020%	-0.052%

READING

4 of 5 assets show a negative or decaying clean MR_REV return at the 20-bar horizon (only USOIL doesn't, and its own 5-bar return is negative). Sample sizes are small per asset (12-36) so no single number here should be treated as settled, but the **direction of the pattern is consistent**: once pre-birth contamination is removed, the clean proof signal does not show a reliable positive edge in this data. That's a materially different picture from what the polluted, unfiltered numbers implied.

Where an asset actually diverges: GBPAUD's 8.2 bucketing

ASSET	WITHIN WINDOW	LATE CLOSE
GOLD	71.3%	28.7%
BTC	70.6%	29.4%
USOIL	63.6%	36.4%
USTEC	66.1%	33.9%
GBPAUD	41.2%	58.8%

GBPAUD is the one place the pattern breaks from the others — **the majority of its eventual retest-to-close proofs arrive late**, well past the 8-candle window, versus roughly a third elsewhere. Worth a dedicated look at GBPAUD specifically before assuming the ~30% late-close rate from the other four generalizes to every FX pair.

What's still open

- **Bucket A for 8.2** (proofs that never close at all) needs a dedicated pass — the reference-level memory used here tracks resolution but not clean "abandoned" detection.
- **GBPAUD's late-close divergence** is unexplained — worth a closer look before assuming the other four assets' ~30% rate generalizes across FX.
- Sample sizes for clean (non-tainted) proofs are small per asset (12-37) — real enough to see a consistent direction, not enough to treat any single asset's number as settled.
- All six items in **Section 10** remain entirely yours to decide — nothing here recommends an answer on any of them, only supplies the data behind each.
- Nothing beyond the logging itself was changed — no thresholds, no gate logic, no live behavior.

Findings & Investigation Plan — reply · GOLD/BTC/USOIL/USTEC/GBPAUD backtests · commits [998bb88](#),

[cbfd7d0](#), [ee73634](#) · All instrumentation is observation-only, gated behind nothing, changes no live behavior.